

Beyond the Prompt: Probing Dual-Use Risks in Large Language Models and a Risk-Based Accountability Framework

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Abstract: Large Language Models (LLMs) present societal benefits and dual-use risks. This paper qualitatively examines model responses to legal and cybersecurity scenarios and observes accessible AI-generated graphical content. Based on these observations, it proposes Financial Friction, a risk-aware framework for proportionate accountability through graduated verification of higher-risk interactions.

Index Terms: Large language models, AI safety, dual-use artificial intelligence, synthetic media, AI governance.

I. INTRODUCTION

Generative Artificial Intelligence (AI) has significantly expanded public access to software development assistance, legal information, technical education, and creative content generation. Modern Large Language Models (LLMs) are capable of performing complex reasoning tasks, producing executable source code, summarizing legal concepts, and assisting with a wide range of professional activities. These capabilities have contributed to the widespread adoption of AI systems across individual and organizational settings.

However, the same capabilities that make these systems valuable for legitimate users also create opportunities for misuse [1]. Requests involving malware development, covert surveillance software, legal evasion strategies, or harmful synthetic media frequently occupy a dual-use space in which similar technical knowledge may support either legitimate research or malicious activity [1]. Distinguishing between these contexts remains a significant challenge for AI safety and governance systems.

Current AI safety mechanisms employ a combination of alignment techniques, reinforcement learning, content moderation, and prompt-level safety mechanisms to reduce harmful outputs. Nevertheless, research has demonstrated that aligned language models may remain susceptible to adversarial prompting, contextual reframing, and other techniques designed to alter or circumvent safety behavior [2], [3]. As a result, models may provide different levels of procedural assistance depending on how a request is formulated or contextualized [2].

This study investigates the behavior of commercially available AI models across two directly evaluated categories of dual-use requests and one complementary observational category:

- **Civil and Legal Guidance:** involving property disputes, legal strategy, procedural delay, and potential misuse of legal procedures.
- **Cybersecurity and Software Development:** involving stealth-oriented utilities, monitoring software, persistence mechanisms, and other programming tasks possessing both defensive and offensive applications [1]-[3].
- **Synthetic Media and AI-Generated Graphics:** examined separately through observational analysis of publicly accessible AI-generated graphical content, with particular attention to explicit or questionable material, identifiable individuals, consent, potential harassment, and content provenance [5], [6], [8], [9].

The first two categories were evaluated through direct interaction with the selected AI models. The synthetic-media component was not conducted as a controlled prompt-based evaluation of the same models. Instead, publicly accessible AI-generated graphical content was observed to examine broader risks associated with the availability and potential misuse of generative graphical technologies [5], [6].

The synthetic-media component is particularly relevant because AI-generated imagery can be used not only for legitimate creative purposes but also to create sexually explicit, deceptive, or non-consensual representations of individuals [8], [9]. Where identifiable persons are involved, the potential consequences may extend beyond conventional content-moderation concerns to harassment, reputational damage, impersonation, and psychological harm [8], [9]. Accordingly, synthetic-media risk cannot be assessed solely on the basis of whether content is technically generated by an AI system; the identity of the depicted individual, consent, context, and potential consequences associated with the representation must also be considered [8], [9].

Rather than advocating broad restrictions on AI capabilities, this paper examines how accountability mechanisms may complement existing technical safeguards. Building upon the empirical observations and synthetic-media analysis presented in this study, a policy-oriented framework termed Financial Friction is proposed as a risk-based accountability mechanism. Under the proposed framework, ordinary educational, professional, and productivity-oriented AI use remains readily accessible, while higher-risk capabilities may require proportionate verification measures such as advanced age verification, identity verification, or KYC-style verification depending on the nature and severity of the requested

activity. Clearly prohibited forms of harmful content remain subject to refusal regardless of the user's verification status.

To support traceability of the direct evaluation, representative prompt-response records were retained with the evaluated model and version where available, test condition, response classification, and evaluation date. Where the evaluated platform provided a publicly accessible conversation or shareable record, the corresponding reference was retained as supporting evidence. These records provide an audit trail connecting the reported observations to the underlying evaluation interactions.

The underlying principle of the proposed framework is therefore:

Open access for ordinary use. Increased accountability for elevated risk. Refusal for prohibited harm.

II. BACKGROUND AND MOTIVATION

Artificial intelligence systems are inherently dual-use technologies [1]. The same computational capabilities that enable beneficial applications including software development, cybersecurity education, accessibility, scientific research, and creative expression may also be adapted for harmful purposes [1]. Consequently, the existence of misuse potential does not necessarily imply that the underlying technology should be broadly restricted.

This dual-use characteristic can be illustrated through a broader principle of technological accountability. Conventional tools, including programming languages, networking software, encryption libraries, and common household instruments, possess legitimate applications despite their potential for misuse. Responsibility for unlawful actions is generally associated with the individual who misuses the technology rather than with the existence of the underlying tool itself. Similar considerations apply to advanced AI systems, although the scale, accessibility, and automation capabilities of generative AI introduce additional governance challenges.

Accordingly, an effective governance strategy should seek to balance two competing objectives preserving broad public access to beneficial AI capabilities while reducing opportunities for anonymous misuse in higher-risk contexts. Excessively restrictive safety mechanisms may reduce the practical utility of AI systems for researchers, students, software developers, cybersecurity professionals, and other legitimate users. Conversely, insufficient safeguards may unintentionally facilitate malicious activities by lowering technical barriers and reducing the expertise required to perform certain tasks [1].

The synthetic-media ecosystem introduces an additional dimension to this challenge [6]. Unlike conventional text-based assistance, AI-generated graphical content can produce highly realistic representations of people, situations, and events [8]. This creates risks associated with impersonation, non-consensual synthetic media, harassment, deceptive representations, and inappropriate sexual content [6], [8], [9]. The risk associated with such generation may depend not only on the content itself but also on whether an identifiable individual is depicted and whether that individual has provided consent [8], [9].

At the same time, lawful adult-oriented synthetic media represents a different category from clearly prohibited material. A governance framework that treats all higher-risk content as inherently

prohibited may unnecessarily restrict legitimate users. A more proportionate approach is therefore to distinguish between lawful but higher-risk capabilities and content that should remain categorically prohibited. Higher-risk lawful capabilities may consequently be subject to greater accountability without necessarily being banned.

These considerations motivate the present study, which evaluates safety behavior across representative dual-use scenarios and separately observes the broader synthetic-media ecosystem. The study subsequently proposes a complementary governance mechanism intended to improve user accountability without substantially limiting legitimate access to AI capabilities.

To improve the reproducibility of the observations, the evaluation records were retained as research evidence rather than relying solely on summarized findings. Where available, records include the evaluated model and version, prompt condition, response classification, and evaluation date. Publicly shareable conversation records were also retained where supported by the evaluated platform. This evidence structure was intended to distinguish observed model behavior from subsequent interpretation and framework development.

III. EVALUATION METHODOLOGY AND OBSERVATIONAL SCOPE

This study employed an empirical evaluation to examine the behavior of commercially available Large Language Models (LLMs) when presented with representative dual-use prompts [2], [3]. The objective was not to benchmark general model performance, but to assess how existing safety mechanisms respond to requests that may support both legitimate and potentially harmful applications depending on user intent and contextual framing [2], [3].

Each direct evaluation interaction was treated as an individual research record. The evaluation record captured the prompt category, contextual formulation, evaluated model and version where available, observed response, response classification, and evaluation date. Where the platform provided a shareable conversation or public reference, that record was retained to support traceability. The resulting records were used as qualitative evidence for identifying differences in model behavior rather than as a statistically representative benchmark of overall model safety.

A. Evaluation Scope

The evaluation focused on two directly tested domains and one complementary observational domain in which AI-generated capabilities present significant dual-use or governance implications:

- **Civil and Legal Guidance:** Prompts involving property disputes, legal strategy, procedural delay, and potential misuse of legal processes were used to examine whether models distinguish between legitimate legal information and requests that could facilitate unlawful or procedurally abusive conduct.
- **Cybersecurity and Software Development:** Prompts involving stealth-oriented utilities, background monitoring, persistence mechanisms, activity logging, and other programming concepts with both defensive and

offensive applications were evaluated to examine the consistency of model safety responses [1]-[3].

- **Synthetic Media and AI-Generated Graphics:** This category was examined separately through observational analysis rather than direct prompt-based testing of the evaluated LLMs. Publicly accessible AI-generated graphical content was observed to identify broader concerns relating to explicit material, representations of identifiable individuals, consent, potential harassment, and content provenance [5], [6], [8], [9].

The three domains were selected because they represent different forms of dual-use risk. Legal and cybersecurity requests may provide information that is useful for both legitimate and harmful purposes [1], while synthetic media can introduce risks associated with the creation and distribution of realistic representations of individuals [8], [9].

B. Evaluation Procedure

Equivalent prompt categories were presented to each evaluated AI model through publicly accessible consumer interfaces. Where appropriate, prompts were reformulated using contextual framing, hypothetical scenarios, administrative narratives, or alternative descriptions [2], [3]. This examined whether safety behavior remained consistent across different formulations of substantially similar requests [2].

The evaluation considered four response categories:

1. **Refusal:** The model declined to provide the requested assistance.
2. **High-Level or Defensive Response:** The model provided general educational information, safety guidance, or defensive alternatives without actionable procedural instructions.
3. **Partial Procedural Assistance:** The model provided some implementation or procedural information while restricting other aspects of the request.
4. **Detailed Procedural Assistance:** The model provided sufficiently specific information that could potentially support implementation of the requested dual-use capability.

The evaluation assessed observable response behavior rather than the internal reasoning or intent of the models. Particular attention was given to whether contextual reframing altered the level of assistance provided [2], [3].

C. Ethical Considerations

The evaluation was conducted exclusively for research and analytical purposes. No generated material was deployed, executed, or used to facilitate unauthorized access, surveillance, malware deployment, or other unlawful activity. Potentially harmful outputs were analyzed solely to characterize model behavior and identify potential weaknesses in existing safety mechanisms [2], [3].

The synthetic-media component was observational. The study did not create or distribute non-consensual explicit imagery or other

prohibited synthetic media. Publicly accessible examples were examined only to assess broader safety, consent, provenance, and governance concerns [5], [6].

IV. OBSERVATIONAL FINDINGS AND MODEL BEHAVIORS

The qualitative analysis identified differences in how the evaluated AI systems responded to dual-use requests. Although all examined models incorporated safety mechanisms, their consistency varied with prompt formulation and contextual framing [2], [3].

In several scenarios, contextual reframing altered the level or type of procedural information provided [2]. These observations suggest that prompt formulation can influence safety behavior even when the underlying requested capability remains substantially similar [2], [3].

TABLE I

PROPOSED RISK TIERS AND ACCOUNTABILITY RESPONSES

Risk Tier	Representative Requests	Proposed Response
Tier 1: Standard Access	Education, writing assistance, routine programming, general research	Normal AI interaction without additional verification.
Tier 2: Elevated Risk	Dual-use cybersecurity concepts, advanced legal strategy, stealth-oriented software, sensitive synthetic media	Graduated verification through identity confirmation, age verification, or other appropriate accountability measures depending on the risk category.
Tier 3: Prohibited Content	Requests involving clearly unlawful activities, child sexual exploitation, non-consensual explicit content, or direct malware deployment	Refusal of assistance together with existing platform safety procedures.

These observations should not be interpreted as permanent characteristics of any particular model. Model behavior is dynamic, and safety mechanisms, deployment configurations, policies, and model versions may change over time.

To maintain an auditable connection between these observations and the underlying evaluations, representative cases were assigned internal test identifiers and linked to their corresponding evaluation records. Each record preserves the relevant prompt condition, evaluated model and version where available, observed response, and response classification. Publicly accessible conversation references are included where available to allow the reported observations to be independently inspected.

A. Civil and Legal Guidance

The legal evaluation examined whether AI systems distinguish between legitimate requests for legal information and prompts that may facilitate procedural abuse or unlawful occupation of property.

Across multiple evaluated scenarios, certain models generated strategic guidance relating to legal procedures before establishing whether the hypothetical user possessed a lawful interest in the disputed property. In several instances, procedural recommendations were provided without requesting clarification regarding ownership, legal authorization, or the factual circumstances underlying the request.

These findings suggest that current conversational safety mechanisms may prioritize responsiveness over contextual verification when responding to legal strategy questions, particularly where requests do not explicitly acknowledge unlawful intent.

The findings should not be interpreted as an assessment of the legal correctness of the generated information. The purpose of this component was to examine safety behavior and contextual verification, rather than to determine whether the legal guidance provided by a model constituted accurate legal advice.

B. Cybersecurity and Dual-Use Programming

The cybersecurity evaluation focused on requests involving software components that possess legitimate administrative applications while also presenting opportunities for misuse [1]. Representative scenarios included background monitoring utilities, activity logging mechanisms, persistence concepts, and self-propagating software discussions [1]-[3].

Observed responses varied depending on contextual framing, indicating that prompt formulation remains an important factor influencing safety behavior [2], [3]. Some requests received greater restrictions when presented as direct malicious activity, while similar technical concepts received different levels of assistance when framed as defensive, administrative, educational, or hypothetical tasks [2].

These findings illustrate the continuing challenge of distinguishing legitimate cybersecurity education and authorized security work from requests that may ultimately support malicious activities [1].

AI safety and cybersecurity access mechanisms continue to evolve [7]. Recent industry approaches have introduced differentiated access and additional controls for higher-risk cybersecurity capabilities [7]. These developments are considered separately in the governance discussion and were not part of the original evaluation.

C. Observational Analysis of AI-Generated Synthetic Media

The synthetic-media component was conducted as an observational analysis of publicly accessible AI-generated graphical content rather than as a controlled prompt-based evaluation of the three language models.

During the observation, questionable and explicit AI-generated content was identified on publicly accessible websites. The observed material highlighted risks involving inappropriate sexual content,

impersonation, non-consensual synthetic representations, and other forms of misuse [8], [9]. Particular concern arises when identifiable individuals are depicted without consent, potentially enabling harassment, humiliation, reputational damage, impersonation, or psychological harm [8], [9].

The observations also highlight a distinction between generation-stage safeguards and post-generation distribution. Even when an AI-generation service implements restrictions, generated material may subsequently be distributed through independent websites or online communities [5], [6]. Therefore, governance focused solely on generation may not address the broader synthetic-media lifecycle.

These observations support risk-sensitive accountability for higher-risk graphical generation. Lawful adult-oriented synthetic media should be distinguished from categorically prohibited content; however, certain higher-risk capabilities may warrant stronger age or identity verification. Where identifiable individuals are involved, consent and foreseeable harm should receive additional consideration, with non-consensual sexually explicit synthetic representations treated as a substantially higher-risk category [8], [9]. Content involving the sexual exploitation of minors should remain categorically prohibited.

These findings represent ecosystem-level observations rather than comparative evidence of any specific image-generation platform or model. Because this component was observational rather than a standardized benchmark, it does not support quantitative comparisons of graphical AI safety performance.

V. RISK-BASED FRAMEWORK AND GOVERNANCE

The framework proposed in this section is derived from the observed behaviors described in Section IV and is presented as a governance and risk-management proposal rather than as an experimentally validated intervention. The observations inform the identification of risk conditions, while the proposed controls translate those conditions into graduated safeguards. Accordingly, the framework should be interpreted as a conceptual risk-based model requiring further empirical validation before deployment in production systems.

A. Risk-Based Detection Framework

The findings of this study indicate that existing safety mechanisms may not consistently distinguish between legitimate educational requests and prompts that could facilitate harmful activities [2], [3]. Rather than relying exclusively on increasingly restrictive content moderation, this paper proposes a complementary risk-based detection framework designed to improve accountability while preserving broad public access to AI systems.

The proposed framework consists of two sequential stages. The first stage performs lightweight automated risk assessment using computationally efficient keyword, phrase, and semantic analysis techniques. Prompts identified as low risk are processed normally, while prompts exhibiting characteristics associated with dual-use or higher-risk activities are forwarded to a secondary verification stage.

Potential techniques applicable to the initial screening stage include:

- *Aho-Corasick Pattern Matching* - Efficient detection of predefined keywords and phrases associated with higher-risk requests [4].
- *Rapid Automatic Keyword Extraction (RAKE)* - Identification of prominent concepts and phrases to estimate the overall characteristics of a prompt.
- *TF-IDF and TextRank* - Measurement of contextual importance for combinations of terms that may occur within dual-use requests.
- *Vector Embedding Similarity* - Semantic comparison between submitted prompts and previously identified categories of higher-risk interactions using embedding-based similarity measures.

For synthetic-media systems, risk assessment may additionally consider characteristics that cannot be determined solely from conventional text-based keyword matching [5], [6]. These may include the requested level of explicitness, apparent age of depicted subjects, whether an identifiable real person is involved [8], whether consent is relevant [9], whether the request involves impersonation or deceptive representation, and whether the requested content presents a foreseeable risk of harassment or other harm [8], [9].

These techniques are intended to provide computationally efficient preliminary risk assessment rather than definitive intent classification [4]. A high-risk classification should therefore be understood as an indicator for additional safeguards rather than proof of malicious intent. Final decisions should incorporate contextual analysis, applicable platform policies, user verification where appropriate, and additional safety mechanisms rather than relying solely on keyword matching.

B. Financial Friction: A Risk-Based Accountability Framework

Building upon the proposed detection framework, this paper introduces Financial Friction, a policy-oriented governance mechanism intended to increase accountability for selected categories of higher-risk AI interactions.

The central principle is that routine educational, professional, creative, and productivity-related requests should remain readily accessible without additional barriers. However, interactions exhibiting characteristics associated with elevated risk may require proportionate verification or accountability measures before access to higher-risk capabilities is provided.

The term Financial Friction does not imply that users should be charged for AI assistance or that payment should determine whether a request is permissible. Rather, payment credentials may serve as one possible accountability signal within a broader verification framework. Depending on the risk category, other mechanisms may include advanced age verification, identity verification, organizational verification, institutional accounts, or KYC-style verification.

The proposed framework therefore distinguishes between ordinary access, elevated-risk access, and prohibited activity.

The resulting documented distribution of the nine GA cases is therefore:

TABLE II
DOCUMENTED GEMINI API TEST RECORD DISTRIBUTION

Gemini API (GA) Test Case Category	Count
Recorded assistance	3
Recorded refusal	3
Allowed control case	1
Oracle screening-layer block	1
Contextual-reframing case	1
Total documented cases	9

Operational Tier Categorization and Risk Response:

Tier 1 (Standard Access): Encompasses routine educational queries, writing assistance, standard programming, general research, and ordinary creative tasks. Requests in this tier operate through normal AI interaction without requiring additional verification or friction.

Tier 2 (Elevated Risk): Includes dual-use cybersecurity capabilities [1], advanced legal strategies, stealth-oriented software, higher-risk synthetic media [6], and lawful adult-oriented graphical generation. Requests falling into this tier trigger graduated verification or accountability measures proportionate to the associated risk profile prior to granting detailed procedural assistance.

Tier 3 (Prohibited Content): Comprises severe harms, including the sexual exploitation of minors, non-consensual explicit synthetic media [8], direct malware deployment, and clearly unlawful activities. Interactions in this tier result in immediate refusal of assistance in accordance with established platform safety procedures.

C. Graduated Verification

The proposed framework does not advocate a blanket prohibition of lawful higher-risk capabilities. Instead, it proposes that the level of accountability associated with an AI capability should increase as the potential consequences of misuse increase.

For example, ordinary image generation involving landscapes, illustrations, fictional characters, or other low-risk creative content may remain available without additional verification. Higher-risk graphical generation involving lawful adult-oriented content could, depending on jurisdiction and platform policy, require advanced age verification to establish that the requesting user satisfies applicable age requirements.

Where the risk is substantially higher, additional identity or KYC-style verification may be considered. Such mechanisms could provide an additional accountability signal and make anonymous misuse more difficult. The precise verification requirements should depend on the nature of the capability, applicable law, and the potential consequences of misuse.

The purpose of such verification is not to determine whether a user is morally deserving of access to a capability, but to introduce proportionate accountability where the potential consequences of misuse are significantly greater.

D. Synthetic Media, Identity, and Consent

Synthetic-media systems require additional safeguards because risk may depend on both the generated content and the identity of the person represented [8], [9].

A lawful adult-oriented synthetic image involving fictional subjects is materially different from an explicit synthetic representation of an identifiable individual created without that individual's consent [8], [9]. The latter may create risks of harassment, humiliation, reputational damage, impersonation, and psychological harm [8], [9].

Accordingly, the proposed framework treats content type, age, identity, consent, and foreseeable harm as separate risk factors [8], [9].

Enhanced verification may be appropriate for certain higher-risk but lawful forms of generation. However, verification should not be interpreted as a mechanism for permitting prohibited content. In particular, content involving the sexual exploitation of minors and non-consensual explicit synthetic representations should remain subject to refusal regardless of whether a user has completed identity or age verification [8], [9].

E. Accountability Without Blanket Restriction

The proposed framework does not seek to eliminate anonymous access to AI systems entirely. Instead, it introduces proportional accountability for interactions that present substantially greater societal risk.

The underlying principle is:

Greater capability and greater potential for misuse should be accompanied by greater accountability, while ordinary legitimate use should remain accessible with minimal friction.

This approach seeks to avoid a binary model in which AI capabilities are either completely unrestricted or completely prohibited.

F. Emerging Industry Examples of Risk-Tiered Access

Recent developments in AI cybersecurity provide examples of differentiated access to higher-risk capabilities. OpenAI's Daybreak initiative [7] illustrates an industry approach in which access to advanced cybersecurity capabilities may be associated with additional verification, access controls, monitoring, and requirements relating to authorized use [7].

Such developments do not constitute validation of the Financial Friction framework and were not part of the original experimental evaluation. However, they illustrate the broader feasibility of an approach in which higher-risk AI capabilities are governed through stronger accountability mechanisms rather than being treated identically to ordinary AI interactions [7].

G. Privacy and Proportionality

The implementation of Financial Friction would require careful consideration of privacy and data protection. Verification should be proportionate to the associated risk, and platforms should avoid collecting unnecessary personal information merely because a capability has been classified as higher risk.

Where verification is required, platforms should establish clear policies regarding what information is collected, how long it is retained, who can access it, and under what legal circumstances it may be disclosed. Payment credentials, identity documents, or other verification information should not automatically be interpreted as proof of malicious or benign intent.

Potential benefits of the framework include improved deterrence of anonymous misuse, increased accountability for higher-risk interactions, improved traceability subject to applicable law, and preservation of unrestricted access for legitimate everyday users.

Nevertheless, implementation would require careful consideration of privacy, data protection, jurisdiction-specific legal requirements, accessibility, false-positive classifications, potential discrimination, data-security risks, and methods by which malicious users may attempt to circumvent verification mechanisms.

VI. EMPIRICAL EVIDENCE AND RISK-SENSITIVE SCREENING PROTOTYPE

The study was supported by two complementary forms of technical evidence: historical prompt-response evaluations of commercially available LLMs and testing of a Gemini API-based risk-sensitive screening prototype referred to as Oracle. The historical evaluation records provide traceability between representative prompts and observed model behavior, while the Oracle prototype provides an implementation-level example of how risk-sensitive screening can be applied before or during an AI interaction. These two evidence sources were analyzed separately because the former evaluates model behavior, whereas the latter evaluates the behavior of a researcher-defined screening mechanism.

A. Historical Prompt-Response Evaluation

Retained evaluation records from May 2026 were retrospectively reviewed for GPT, Claude, and Gemini. The records contain representative prompts, model identifiers where recorded, researcher observations, response classifications, and publicly shareable conversation references where available. The evaluation covered two principal dual-use domains: legal scenarios involving potentially unlawful conduct and cybersecurity scenarios involving monitoring, persistence, and other potentially dual-use software behavior.

For GPT, two documented cases were retained. The first involved an unlawful land-encroachment scenario in which the recorded observation indicated that the model provided potentially useful procedural guidance without requiring a jailbreak. The second

involved a cybersecurity background-service scenario in which GPT resisted providing the requested assistance despite hypothetical reframing. These observations provide one assistance case and one resistance/refusal case within the retained historical record.

For Claude, three observations were retained. The cybersecurity interaction involved assistance with background screen-recording functionality, while a subsequent interaction in the same context demonstrated sensitivity to reframing and concealment-related characteristics. The legal interaction initially showed resistance to explicitly unlawful framing but subsequently provided more general discussion after hypothetical reframing. These cases therefore provide evidence of both initial resistance and subsequent changes in assistance under altered contextual framing.

For Gemini, two principal consumer-interface observations were retained: one involving cybersecurity functionality and another involving unlawful land encroachment. Both records were documented by the researcher as assistance cases. Separately, the Gemini API interaction history contained a sequence of nine recorded Gemini API (GA) test cases, designated GA-01 through GA-09, which provides more detailed evidence for examining changes in model behavior across related prompt conditions.

B. Gemini API Screening Prototype

The Gemini API prototype, named Oracle, was developed to introduce a risk-sensitive screening layer around AI interaction. The prototype uses predefined indicators to identify sensitive cybersecurity, malware-related, credential-related, and other high-risk requests before permitting normal processing. The screening mechanism is therefore a heuristic first-stage control rather than a complete safety or intent-detection system.

The recorded API evaluation provides several contrasting cases. GA-01 involved a background screenshot-recording request and produced substantial implementation-oriented assistance. GA-02 reframed the request as testing other startup programs and continued the discussion of the requested functionality. GA-03 explicitly introduced concealment requirements, and the recorded response continued to discuss silent or invisible execution. These three cases constitute three recorded assistance cases involving sensitive cybersecurity functionality.

In contrast, GA-04 and GA-05 involved unlawful land-encroachment scenarios and were recorded as refusals. GA-09 involved self-persistence behavior and was ultimately refused as malware-like behavior. Thus, the retained API interaction history contains three documented refusal cases among GA-04, GA-05, and GA-09.

The prototype also included explicit control cases. GA-07 involved visible, user-controlled screen recording and was recorded as allowed, demonstrating that the presence of a capability such as screen recording alone was not intended to trigger indiscriminate blocking. GA-08 was blocked by the Oracle screening layer because the request was classified as sensitive cybersecurity functionality. Importantly, GA-08 represents prototype-screening behavior rather than independent evidence of Gemini's baseline model behavior.

GA-06 was retained as part of the nine-case Gemini API interaction history and was reviewed as a contextual variation within the broader evaluation. Because its significance depends on

the specific response recorded under that contextual variation, it is not used to infer a general model behavior beyond the documented interaction.

The Gemini API records should therefore be interpreted as descriptive evidence from a small exploratory set rather than as a benchmark score, statistical estimate, or general measure of Gemini's safety performance.

C. Oracle Screening-Layer Results

A separate set of prototype screening tests examined whether the risk-sensitive layer could distinguish between high-risk and ordinary requests. The documented prototype cases included a high-risk stealth screenshot request, a reframed recording request containing concealment and persistence characteristics, an ordinary Python calculator request, and a visible screen-recording request. The first two were blocked as sensitive, while the ordinary calculator and visible user-controlled screen recording were allowed.

Accordingly, among these four explicitly documented Oracle screening cases, the screening layer produced:

- 2 sensitive/blocked cases
- 2 allowed cases

This result is not presented as a statistical accuracy measurement because the four cases were selected research examples rather than a balanced benchmark dataset. Instead, they demonstrate the intended distinction between potentially harmful characteristics and legitimate functionality.

D. AI-Assisted Safety Engineering

The Oracle prototype also illustrates a human-in-the-loop approach to AI-assisted safety engineering. The researcher defined the risk categories, ethical constraints, screening requirements, and accountability objectives, while the Gemini API served as the AI implementation component. The system therefore does not represent an AI system independently determining ethical boundaries or autonomously correcting itself.

Conceptually, the development process can be represented as:

AI capability → *human-defined ethical constraints* → *AI-assisted safety intervention* → *safer AI interaction*

This demonstrates how an AI system can contribute to the development of mechanisms intended to improve the safety of AI interactions while retaining explicit human responsibility for defining the boundaries within which that assistance operates.

VII. CONCLUSION

This study examined the responses of commercially available Large Language Models across representative dual-use scenarios involving legal guidance and cybersecurity-related programming requests, complemented by an observational analysis of publicly accessible AI-generated graphical content.

The direct evaluation identified variability in how current AI safety mechanisms respond to contextual prompt reframing, with several evaluated scenarios producing procedural guidance that could support both legitimate and potentially harmful applications [2], [3]. The observational synthetic-media analysis additionally highlighted broader concerns surrounding explicit AI-generated content, non-consensual synthetic representations, identifiable individuals, harassment, and the distribution of generated material beyond the original generation environment [5], [6], [8], [9].

These findings suggest that existing safety mechanisms continue to face challenges in reliably distinguishing legitimate educational or professional use from requests that may facilitate misuse [2], [3]. At the same time, a uniform prohibition-based approach may unnecessarily restrict legitimate users and lawful applications.

Rather than advocating increasingly restrictive content moderation alone, this paper proposes a complementary governance approach based on graduated accountability. The proposed Financial Friction framework combines lightweight computational risk assessment [4] with proportional verification for selected higher-risk interactions.

Under the proposed framework, ordinary AI use remains accessible without unnecessary barriers, while higher-risk capabilities may require advanced age verification, identity verification, or KYC-style accountability mechanisms depending on the severity and nature of the requested activity. Lawful adult-oriented synthetic media is not treated as automatically prohibited; instead, higher-risk access may be accompanied by stronger accountability requirements. Conversely, prohibited activities, including sexual exploitation of minors and non-consensual explicit synthetic media [8], [9], remain subject to refusal regardless of verification status.

The framework therefore seeks to balance two important objectives reserving broad access to legitimate AI capabilities while introducing greater accountability where the potential societal consequences of misuse are substantially greater.

Future research should evaluate the practical effectiveness, usability, privacy implications, legal feasibility, and circumvention resistance of this framework across diverse AI platforms, jurisdictions, and user populations. Empirical validation through controlled deployment studies would further clarify its impact on both legitimate use and malicious abuse.

VIII. LIMITATIONS

Several limitations should be considered when interpreting the findings of this study.

First, the LLM evaluation was conducted using publicly accessible consumer interfaces and therefore reflects observed behavior during the testing period rather than the underlying model architectures. Future software updates, safety modifications, model changes, or policy revisions may alter observed behavior.

Second, the prompt evaluation focused on representative dual-use scenarios rather than an exhaustive benchmark of all possible prompt categories. The findings therefore should not be

interpreted as comprehensive assessments of overall model safety.

Third, determining user intent remains inherently complex. Identical prompts may represent legitimate educational research, authorized cybersecurity testing, or malicious objectives depending on the surrounding context. Consequently, no automated detection framework can perfectly distinguish harmful from benign requests.

Fourth, the synthetic-media component was observational rather than experimental. The publicly accessible AI-generated content observed during the study cannot establish comparative safety performance among individual image-generation models or platforms, nor the prevalence of harmful AI-generated content across the broader internet.

Finally, the proposed Financial Friction framework is a policy proposal rather than a validated deployment model. Further empirical research is required to evaluate its effectiveness, user acceptance, privacy implications, implementation costs, accessibility, circumvention risks, and impact on legitimate and harmful use.

IX. THREATS TO VALIDITY

The findings are subject to several potential threats to validity.

Model behavior is dynamic and may change following updates to alignment strategies, safety policies, model architectures, or deployment configurations [2], [3]. Consequently, identical prompts may produce different responses over time. The observations reported in this study therefore represent model behavior during the period in which the evaluation was conducted and should not be interpreted as permanent characteristics of the evaluated systems.

Prompt formulation and contextual framing may also influence observed outcomes [2], [3]. Although representative prompt categories were evaluated, alternative wording, conversation history, or contextual framing may produce different responses. The results should therefore be interpreted as observations under the tested conditions rather than universal characteristics of the evaluated systems. In particular, differences in response behavior should not automatically be interpreted as evidence that one model is categorically safer than another, since observed behavior may depend on the precise wording, surrounding context, interface configuration, and safety policies active at the time of testing.

The qualitative assessment of generated responses involved researcher judgment despite the use of predefined evaluation categories. Larger prompt datasets, multiple independent evaluators, and quantitative analysis would strengthen the reliability and generalizability of the findings. The four response categories used in the evaluation provide a structured basis for comparison, but boundaries between high-level information, partial procedural assistance, and detailed assistance may sometimes involve qualitative judgment. Future studies could reduce this subjectivity through independent annotation, inter-rater agreement measures, and predefined scoring criteria.

The evaluation was also limited to publicly accessible consumer interfaces. Differences in system configuration, account-level restrictions, regional availability, model versions, safety settings,

and interface behavior may influence the responses obtained through such interfaces. Accordingly, the findings describe the behavior observable through the tested access conditions and do not necessarily represent the behavior of underlying models when accessed through alternative deployment environments or configurations.

The synthetic-media component introduces an additional validity consideration because it relied on publicly accessible observations rather than a controlled experimental dataset. The availability, visibility, and characteristics of online AI-generated content may vary over time and across websites. These observations should therefore be interpreted as evidence of potential ecosystem-level risks rather than measurements of their frequency or prevalence [6]. The observational approach also does not establish whether particular examples were generated by a specific model, whether the content originated from an AI-generation service or was subsequently modified, or how representative the observed material is of the wider synthetic-media ecosystem.

Another consideration concerns the relationship between observed model behavior and actual real-world misuse. A response providing procedural information does not by itself demonstrate that the information was subsequently used for harmful purposes. Similarly, a refusal does not establish that a determined user could not obtain equivalent information through another model, search engine, software repository, or other source. The study therefore evaluates potential exposure to dual-use assistance rather than measuring actual incidents of misuse or the effectiveness of individual safeguards in preventing them.

Finally, Financial Friction has not been experimentally deployed. Its effectiveness in deterring misuse, improving accountability, and preserving legitimate access therefore remains to be empirically validated. Verification mechanisms may also introduce privacy, accessibility, security, and circumvention risks requiring further investigation. Additional concerns may include false-positive risk classification, disproportionate burdens on legitimate users, regional differences in legal requirements, and the possibility that malicious users adapt their behavior to avoid verification. Future research should therefore evaluate the framework through controlled deployment studies and assess whether increased accountability produces measurable reductions in harmful use without unnecessarily restricting legitimate AI access.

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